

Assignment 1 – SPA
Situated Learning Assignment
Domain- Smart Cities & Urban Infrastructure

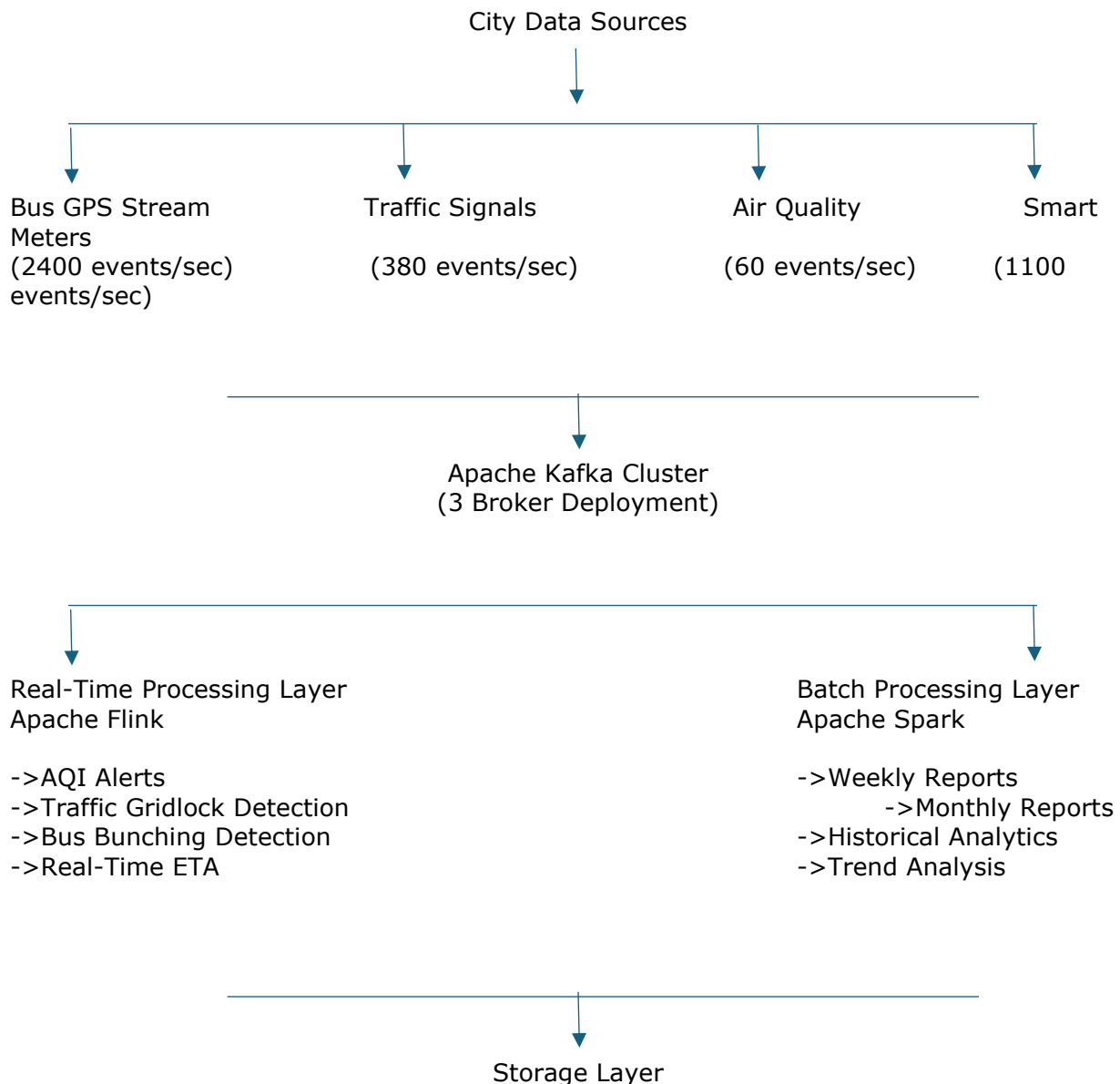
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TASK A - Architecture Design — Lambda vs Kappa for a Dual-Reporting City Platform

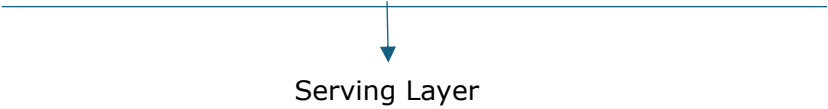
Problem statement 1:

Draw a labelled UrbanPulse architecture diagram showing all four data sources, the Kafka ingestion layer, processing layers (both real-time and batch), storage layer (specify which database technology you would use for: time-series sensor data, geospatial bus positions, historical AQI records, and councillor report aggregates), and the serving layer (dashboards, advisory APIs, signal control interface).

Solution 1:



Data Type	Technology	Purpose	Reason
Sensor Data Bus Positions	InfluxDB PostgreSQL + PostGIS	Time Series Geospatial Queries	High write rate GIS support
AQI Records	Apache Iceberg	Historical Storage	Cheap analytics
Reports	PostgreSQL	Aggregations	Structured govt



- Smart City Dashboard
- AQI Advisory API
- Traffic Control Interface
- Bus ETA Mobile Service
- Councillor Reporting Portal

Problem Statement 2:

Construct a formal Lambda vs Kappa Architecture evaluation matrix for UrbanPulse. Rows: Latency, Fault Tolerance, Operational Complexity, Reprocessing Capability, Cost, Compliance with Government Reporting Mandate. Columns: Lambda, Kappa. Fill each cell with a specific UrbanPulse-grounded assessment — not generic text. Conclude with your architecture choice and justify it.

Solution 2:

Lambda vs Kappa Evaluation Matrix

Criteria	Lambda Architecture	Kappa Architecture
Latency	Very low (< seconds) via speed layer	Very low using Kafka replay
Fault Tolerance	High due to batch recomputation	High via Kafka persistence
Operational Complexity	High (batch + speed pipelines)	Lower (single streaming stack)
Reprocessing Capability	Excellent using batch layer	Good through Kafka replay
Cost	Higher infrastructure cost	Lower infrastructure footprint
Government Reporting Compliance	Excellent for weekly/monthly reports	Possible but requires replay and aggregation jobs

UrbanPulse Assessment

Criteria	Lambda (UrbanPulse)	Kappa (UrbanPulse)
Latency	Supports AQI alerts <2 min and signal adaptation <90 sec	Supports same requirements
Fault Tolerance	Batch layer can rebuild lost city datasets	Relies primarily on Kafka retention
Complexity	Flink + Spark + Batch storage maintenance	Simpler operational model
Reprocessing	Historical AQI records easily recomputed	Reprocess by replaying Kafka topics
Cost	Additional hardware/storage needed	Lower city infrastructure budget
Government Reporting	Strong support for mandatory monthly/state reports	Requires additional aggregation logic

Recommended Choice: Lambda Architecture

Why?

Metro Connect requires:

- Real-time response
- Weekly reports
- Monthly compliance reports
- Historical trend analysis
- Auditability

Since government reporting is a mandatory requirement, Lambda Architecture provides stronger compliance support while still delivering real-time analytics.

Problem Statement 3:

Write an Architecture Readiness Checklist for a government smart city deployment (minimum 12 items). The checklist must address: data sovereignty (data must not leave city servers), open-source mandate, disaster recovery requirements (RPO < 15 min, RTO < 30 min), and accessibility of dashboards for non-technical ward officers.

Solution 3:

#	Checklist Item
1	Data remains within city-owned servers (Data Sovereignty)
2	All components use open-source technologies
3	Kafka cluster configured with 3 brokers
4	Encryption at rest enabled
5	TLS encryption for data in transit
6	Role-based access control implemented
7	RPO less than 15 minutes
8	RTO less than 30 minutes
9	Daily automated backups
10	Multi-zone disaster recovery deployment
11	Real-time monitoring and alerting configured
12	Dashboard accessible for non-technical ward officers
13	ADA/Government accessibility standards met
14	Audit logging enabled
15	Kafka topic retention policies documented
16	Security and privacy review completed

